



Degree Project in Media Technology

First cycle, 15 credits

Artifact Perception in Algorithms for Audio Time Stretching

Uppfattning av artefakter i algoritmer för tidsskaleändringar av ljud

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Abstract

This study explores user preferences and perceptions regarding algorithms used for time-scale modification (TSM) of audio. While features for adjusting playback speed are widely available in media players, the underlying algorithms and their differences are often not well understood by everyday users. Given that different TSM algorithms can produce vastly different audio artifacts, we investigate whether non-expert users exhibit consistent preferences or can articulate qualitative opinions about the results produced by different algorithms.

Our findings may be relevant for software developers faced with a selection of audio stretching algorithms, as understanding user perception of algorithmic artifacts can inform design decisions. Drawing on previous research in TSM techniques and methods for qualitative and quantitative analysis, we provide context for the causes of TSM-related artifacts and use it to interpret user preferences.

We conducted a within-subjects study with 29 participants, who evaluated four TSM algorithms across 16 test conditions, which were combinations of audio types and playback speeds. Rankings were analyzed using the Friedman test and aligned rank transform ANOVA to assess the statistical significance of algorithm preferences. Additionally, participants provided qualitative feedback describing their listening strategies and perceptions.

Results indicate strong agreement among participants on the best-performing algorithm within each tested condition. While no single algorithm consistently outperformed the others across all scenarios, playback rate and audio type significantly influenced which algorithm was preferred. Despite this variability, participants largely converged in their rankings for most conditions. The qualitative analysis revealed recurring themes in artifact perception, including a dislike of fluttering, phasiness and lack of clarity.

Sammanfattning

Denna studie undersöker användarpreferenser och uppfattningar kring algoritmer som används för ändringar i tidsskalan (TSM) av ljud. Funktioner för att justera uppspelningshastighet är vitt tillgängliga i mediaspelare, men de underliggande algoritmerna och deras skillnader är ofta inte väl förstådda av gemene användare. Eftersom olika TSM-algoritmer kan producera mycket olika ljudartefakter, undersöker vi huruvida icke-expertter uppvisar konsekventa preferenser eller kan formulera kvalitativa åsikter om resultaten som fås av olika algoritmer.

Våra resultat kan vara relevanta för mjukvaruutvecklare som står inför valet av algoritm för tidsskaleändringar för ljud, eftersom förståelse för hur användare uppfattar artefakter kan stödja beslut i designprocessen. Med utgångspunkt i tidigare forskning om

TSM-tekniker och metoder för kvalitativ och kvantitativ analys, presenterar vi orsakerna till TSM-relaterade artefakter och använder denna kunskap för att tolka användarpreferenser.

Vi genomförde en inomgruppsstudie med 29 deltagare, som utvärderade fyra TSM-algoritmer över 16 testförhållanden, som utgjordes av kombinationer av ljudtyper och uppspelningshastigheter. Rankningarna analyserades med hjälp av Friedman-testet och *aligned rank transform* ANOVA för att undersöka statistisk signifikans i algoritmpreferenser. Deltagarna gav även kvalitativ återkoppling där de beskrev sina lyssningsstrategier och upplevelser.

Resultaten visar på stark enighet bland deltagarna om vilken algoritm som presterade bäst för varje testförhållande. Ingen enskild algoritm presterade konsekvent bäst i alla scenarier, utan uppspelningshastighet och ljudtyp hade en stor påverkan på vilken algoritm som föredrogs. Trots denna variation tenderade deltagarna att vara överens i sina rankningar i de flesta fall. Den kvalitativa analysen visade återkommande teman i uppfattningen av artefakter, där ogillande av hackighet, "phasing"-fenomen, samt en brist på klarhet i ljudet var vanliga.

CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Human-centered computing**; • **Computing methodologies** → **Perception**;

Keywords

Audio; Perception; Time-Scale Modification; Time Warping; Pitch Shifting; Listening; Sound Manipulation; Sound Artifacts; Playback Speed

1 Introduction

Features for changing playback speed have become common in digital media players. Their use ranges from speeding up lengthy podcast episodes and recorded lectures, to slowing down difficult pieces of music when practicing an instrument. However, simply changing the playback rate of audio brings about a corresponding change in pitch. This may be preferable in some cases — perhaps when altering the tempo of non-tonal material such as drums and percussion — but entirely unsuitable when dealing with harmonic content. Therefore, choosing appropriate algorithms to perform the change in playback rate while preserving pitch is vital in order to retain the character of the original audio.

The task of altering the playback speed of audio without changing its pitch is called *time-scale modification* (TSM) [8], often referred to as *time stretching* or *time warping*. It is applied to audio in many different settings; podcast or audio book listeners may speed up content, musicians slow down difficult pieces of music, language learners slow down speech recordings, and deejays match the tempo of two recordings. Additionally, whenever the playback speed of a video is altered, it is usually desired that the audio playback rate is matched to the video rate.

Different algorithms for TSM introduce different *artifacts* — unwanted effects resulting from an operation on a signal. More complex, high-end algorithms may perform better in preventing unwanted artifacts, but are difficult to implement in real time due to their processing requirements. Conversely, simpler algorithms often introduce more unwanted artifacts, but work flexibly in real time and may still yield acceptable results if the change in playback rate is small.

From a software developer’s perspective, the implementation of a time stretching algorithm must be made with several considerations in mind. What artifacts will the algorithm produce? How high are the computational demands? Will the algorithm function in real time or cause latency? On mobile and tablet devices, features for altering playback speed can be implemented using algorithms that come with the operating systems’ software development kits [2]. Web browsers also feature similar built-in functionality [9, 23, 24].

In applications intended for media playback, it’s often desired that users may change playback speed on the fly with no latency. In settings where audio is being processed rather than simply played back — such as music production or audio editing — algorithms can afford to be more computationally intensive since these applications are not constrained by real-time requirements.

The purpose of this study is to investigate whether any patterns can be identified with regards to user preferences in algorithms for time-scale modification of audio. We do not seek to identify a single “best” algorithm or to draw performance conclusions about the specific algorithms included in our evaluation. Rather, we aim to discover whether consistent preference trends arise when listeners adjust playback speed for various types of audio. To this end, participants completed a survey in which they ranked four different TSM algorithms according to personal preference. Each participant evaluated 16 combinations of four audio types and four playback speeds, providing insight into how algorithm preference varies under these conditions.

The broader topics of audio processing and audio engineering have been widely studied, but only limited research has covered the layperson’s perspective on common audio processing techniques, despite their application in software used daily by many. Understanding user preferences in TSM algorithms may be of importance to developers who are faced with a choice between several algorithms. If preferences tend to converge significantly, this may indicate that an informed developer could safely trust their own taste as it is likely to conform with that of users. If preferences are greatly varied, developers may need to take measures to better understand their users and how to accommodate their preferences.

2 Background

The task of changing the playback rate of audio while preserving pitch can be achieved using a number of different algorithms. They require varying amounts of computational power, and have different sonic characteristics owing to the artifacts they introduce. Some cause what is known as “smearing”, characterized by a loss of transients. Others, a noticeable “choppiness” as changes in playback rate become large.

A main challenge in audio time stretching is preserving different kinds of sonic characteristics. Procedures well-suited to retaining harmonic information, such as sustained musical notes, may be ill-suited to preserving transients and non-periodic sounds, such as clicks and noises. For example, when stretching a recording of a violin played together with castanets, the violin’s pitch and timbre must be maintained, as well as the castanets’ non-tonal clicks and their precise timing. [8]

Pitch-scale modification (or pitch shifting, pitch warping) is a procedure closely related to time-scale modification. It involves changing the pitch of audio signals without affecting their duration. Many of the same algorithms can be used for pitch shifting as are used for time stretching [13, 14].

2.1 Fundamentals of Time-Scale Modification

Featured in nearly all TSM algorithms is some form of overlap-add (OLA) mechanism. It involves dividing an audio signal into short frames of equal size, shifting them forward or backward along the time axis, and joining them back together. The division of the signal into frames preserves local information within each frame, such as frequency and phase relationships, while the time shifting of the frames performs the global time-scale modification.

The distance between the starting points of two consecutive frames, prior to shifting them in time, is called the *analysis hopsize*. After the time shift has been performed, the distance is referred to as the *synthesis hopsize*. The synthesis hopsize is thus a function of the analysis hopsize and the stretch factor. Similarly, the frames themselves are referred to as *analysis frames* and *synthesis frames*. In order to prevent discontinuities in the signal resulting from the rejoining of the time-shifted synthesis frames, the analysis frames are constructed with some amount of overlap. In the overlap of two frames, a crossfade is typically applied using a *Hann window*, resulting in a smoother transition between neighboring synthesis frames. [8]

A common artifact resulting from the basic overlap-add approach is *fluttering*, caused by phase jumps during the transition between

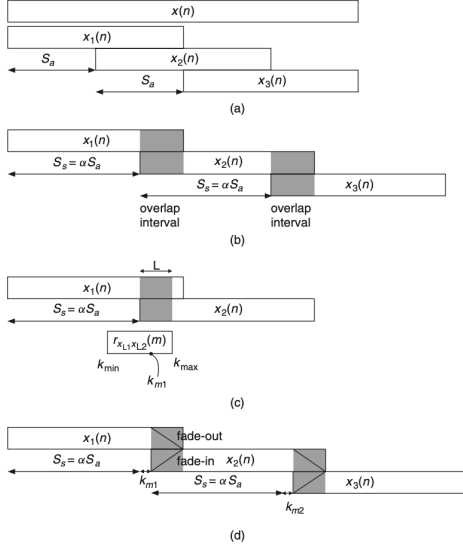


Figure 1: Illustration by Zölzer (2011) visualizing the synchronous overlap-add (SOLA) algorithm. a) Segmentation of input signal, b) repositioning of the blocks, c) computation of cross-correlation and synchronization of the blocks, d) overlap and add. [27]

two neighboring frames. It occurs when the waveforms of the frames at their respective crossover points are not similar, or are out of phase, causing a drop in amplitude at the crossover. This effect is remedied in the *SOLA* (synchronous overlap-add) variation proposed by Roucos and Wilgus [16]. The *SOLA* algorithm uses a variable-size overlap, where the point of transition between two successive frames is determined by the maximum of a cross-correlation function. As a result, the transition between frames occurs where the two are as “similar” as possible, and fluttering artifacts are greatly reduced. The *SOLA* algorithm is illustrated in figure 1.

2.2 Frequency-Domain Operations

The *OLA* and *SOLA* procedures for TSM are constrained to the time domain. By performing the discrete Fourier transform on analysis frames, called a short-time Fourier transform (STFT), an input signal is deconstructed into its constituent frequency components. This produces a representation of the signal in a joint time and frequency domain, consisting of time-successive frequency-domain frames. [14]

The *phase vocoder* algorithm makes use of the short-time Fourier transform and is thus able to produce time-scale modified signals where the phases of the component frequencies between synthesis frames are aligned individually, resulting in an output signal with no phase jumps. However, the per-frequency phase alignment causes a misalignment of frequency components within the same synthesis frame. This results in a loss of transients, or “peaks” in the audio signal — an artifact known as *smearing*. Modern phase vocoders contain countermeasures for reducing smearing, such as peak detection mechanisms ‘resetting’ the phase of individual

frequency components to match those of the analysis frames at the time of a peak [13].

2.3 Algorithms Used in the Test

Four TSM algorithms were selected for evaluation in the survey. They will throughout the text be referred to as *OLA*, *SOLA*, *phase vocoder*, and *élastique*. These specific algorithms were chosen because they produce artifacts with different characteristics and require varying amounts of computational effort. Although they do not come close to covering the full range of TSM techniques, we expected their qualitatively different outputs to produce distinct and meaningful responses from the test participants. The *OLA* and *SOLA* algorithms operate in the time domain, making them suitable for real-time use. The *élastique* and *phase vocoder* algorithms contain more computationally demanding frequency-domain operations, which makes them unsuitable for real-time use.

2.3.1 *OLA*. The *OLA* (overlap-add) algorithm is a basic mechanism featured in most TSM algorithms, described in section 2. The *simple windowed* option featured in digital audio workstation Reaper is based on the *OLA* technique, and offers a choice of window size and crossfade lengths. For this test, we opted for the default settings of 50-millisecond windows with a 50 percent fade. These settings were consistent across all playback speeds and audio types. We used Reaper version 7.34.

2.3.2 *SOLA*. For the *SOLA* algorithm, we used version 3.7.1 of Audacity, a free and open source software for sound recording and editing [4]. Audacity offers two algorithms for time stretching, one simple and one complex. The simple algorithm is based on the *SOLA* technique, and was used for our test. The algorithm itself was provided to Audacity by the SoundTouch library [15].

2.3.3 *Phase Vocoder*. Apple’s music production software Logic Pro version 11.1.2 features the *Flex Time - Polyphonic* algorithm, based on the phase vocoder technique. Being proprietary software, we have no means of investigating the exact properties of the algorithm other than its general implementation of the phase vocoder, as described on Apple’s support page [3]. The ‘complex’ setting was enabled, which is said to improve handling of transients.

2.3.4 *Élastique*. The *élastique 3.3.3 pro* algorithm is proprietary software by zplane used for deejaying, music production, and live performance [26]. We applied the algorithm using Reaper, and used the default settings for the *élastique* processing option. *Élastique* can operate in real time, but requires much processing power as well as an analysis of the entire input signal. As with Logic Pro’s phase vocoder algorithm, we cannot know what precise processes are included in the *élastique* algorithm.

2.4 Previous Work

Research into time-scale modification of audio has mainly focused on the development and refinement of existing algorithms from a computer science and mathematics perspective [7, 11–14, 16, 21]. Additionally, research into perception aspects of audio processing has mainly covered the application of conventional effects, such as distortion, equalization, and compression [5, 6]. However, these

studies have used test groups consisting exclusively of music production students with experience in audio engineering, instead of media consumers with “untrained ears”.

3 Method

The method consisted of the selection of algorithms, playback speeds and audio files, the creation and distribution of a survey, and analysis of the collected data. From an early stage, we decided the survey had to be large enough to accommodate a variety of test conditions, but short enough to avoid causing the participants listening fatigue.

The following sections will detail each step in our process. First, the creation of audio files for use in the survey. Then, the creation of the survey and our distribution strategy. Finally, we will describe the methods used for processing and analyzing the collected data.

3.1 Audio Files

Four eight-second audio clips were selected for the test: a speech recording, a classical music recording, an instrumental band recording, and a musical recording with vocals. The recordings were selected with a variety of spectral and musical properties in mind: the speech recording is a clean, high-fidelity recording of a single voice with no background sounds; the classical music recording is of a polyphonic string orchestra; the instrumental band recording contains drums, electric bass, electric guitar, and keys; and the musical recording with vocals contains singing, guitar and lighter drums.

The speech recording was sourced from the Sveriges Radio program *Bildningsresan: Paris* (29 June 2024).¹ The classical recording is a snippet of Beethoven’s *Symphony No. 5 in C Minor, Op. 67: I. Allegro Con Brio*, performed by the Dresden Philharmonic Orchestra & Herbert Kegel.² The instrumental band recording is *Nightfall* by Mezzoforte.³ The music recording with vocals is Joni Mitchell’s *You Turn Me On I’m A Radio*.⁴

3.1.1 Playback Speeds. Each combination of audio and algorithm was rendered at four different playback speeds (1.5x, 0.9x, 0.75x, 0.5x), for a total of 64 audio files to be used in the test. The selection of playback speeds was made with three main considerations in mind: (1) all four playback speeds are common options in media players; (2) slowing down audio is a more complicated process which introduces more artifacts compared to speeding up audio; (3) choosing more than four speeds made the survey too time consuming for participants.

3.1.2 Other Processing and Rendering. Both prior to applying the time warping and after rendering, each audio file was separately normalized to an integrated loudness of -20 LUFS (loudness units relative to full scale) using Logic Pro’s loudness normalization feature. This step proved important, as one stretching algorithm resulted in around 2 dB of amplitude loss compared to the original audio files. No other processing was applied.

3.2 Survey

The survey was created using the free, web-based platform *Aidaform* [1]. It was designed with six pages, where the first asked participants for personal information, the middle four dealt with each respective audio file, and the last asked participants for follow-up information. Before these six pages was a welcome page, where general information and instructions were presented.

Some questions required participants to write their answers freely, while others required selecting a number on a scale. Some also had a multiple choice format. From now on, free-text questions will be marked with *[text]*, multiple choice with *[multichoice]*, and scale-based questions will be marked with *[scale]*. This distinction is important because the scale responses are intended for quantitative analysis, whereas the text and multiple choice responses will be discussed qualitatively to contextualize the scale rankings. The way we treat scale ranking data is presented in section 3.4. We will now detail the layout of the survey pages.

3.2.1 Initial Questions. The first survey page collected information on demographics, previous experience with audio time stretching, and playback speed habits for media consumption. These were collected before subjecting the participants to any listening. In order to keep the parameters narrow, demographic information was restricted to age and gender. Participants were asked the following specific questions about their background and habits:

- How much experience do you have with audio time stretching? *[multichoice]*
- How bothered are you by poor audio quality in speech? *[scale]*
- How bothered are you by poor audio quality in music? *[scale]*
- Do you ever change the playback speed when consuming audiovisual media such as music, podcasts, movies and TV-shows, or recorded lectures? *[multichoice]*
- If you ever change media playback speed, why? Do you have a particular use for it? *[text]*

3.2.2 Ranking Questions. The four ranking pages of the survey each pertained to one of the four audio files. Every page first presented the original eight-second audio at 1.0x speed for comparison, as shown in figure 2. Then, divided into subsections based on playback speed, four versions of the audio were presented at a time, time-stretched by the different algorithms. Using a ranking matrix, participants were asked to rank the algorithms at the current playback speed according to their preference. Participants were instructed to award tie ranks to algorithms that were not discernibly different. This process was repeated for each playback speed and audio file. Figure 3 shows the layout of a ranking subsection.

In each section, the algorithms were anonymized with labels A, B, C, and D. In order to reduce learning effects, the algorithm labels were shuffled after each audio clip. However, the labels were consistent across the different playback speeds for the same audio clip.

To aid in a more nuanced understanding of the participants’ listening criteria when examining the audio files, each page had a dedicated question at the bottom about what sonic aspects the

¹At 8:30. Recorded from <https://www.sverigesradio.se/avsnitt/bildningsresan-paris>

²At 2:32. Recorded from the collection *Beethoven, Brahms & Others: Orchestral Works* on Apple Music.

³At 0:40. Recorded from the album *Forward Motion* on Apple Music.

⁴At 1:12. Recorded from the album *For the Roses (2022 Remaster)* on Apple Music.

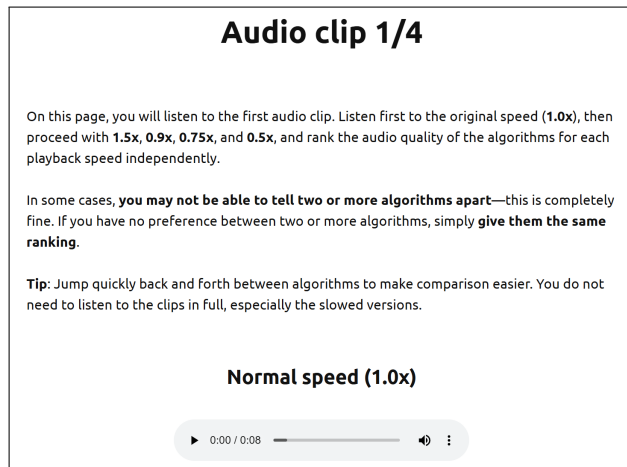


Figure 2: The initial text of a ranking page, followed by the audio at normal speed.

listener had focused on when deciding the ranking order. The questions were not the same on all pages. For both the instrumental song and the song with vocals, we asked what specific instruments participants had focused on and provided multiple answer choices. More than one option was allowed to be selected. For the classical and speech recordings, we instead asked the participants about their listening focus in a free text format.

3.2.3 Closing Questions. To make sure any unpredicted factors didn't interfere with the results, the final page was dedicated to retrieving data on each participant's specific test environment. The specific questions were as follows:

- What type of headphones/speakers did you use in the test? [multichoice]
- Did you take the test in a quiet environment? [multichoice]
- How important do you believe the choice of algorithm for audio stretching is to your listening experience when consuming audiovisual media? [scale]
- When comparing speech to music, which do you think is most sensitive to the choice of stretching algorithm? [multichoice]
- Did you experience a loss of focus or listening fatigue as the test progressed? [multichoice]
- Any closing thoughts? (optional) [text]

3.3 Distribution

We distributed the survey publicly via Slack and Discord channels associated with the Chapter for Media Technology at KTH, as well as directly to friends and family. Since age group and gender was the only demographic information collected, we have no means of identifying individual respondents. As a consequence of our choice of distribution channels, a majority of respondents were predicted to be current or former engineering students aged between 19 and 30 years. The actual age distribution is presented in section 4.

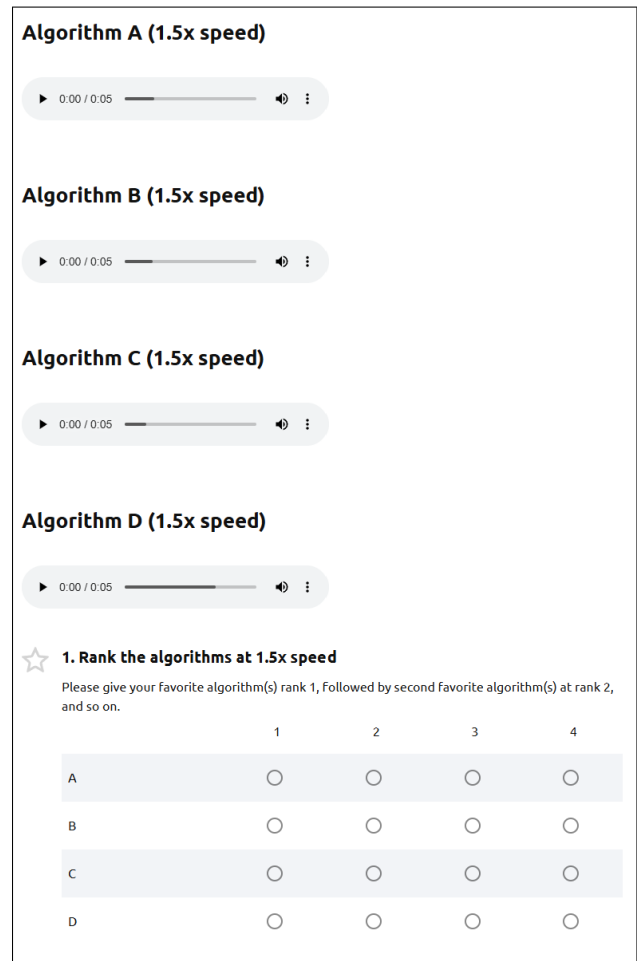


Figure 3: The ranking subsection layout.

3.4 Quantitative Data

The algorithm rankings were stored as numbers between 1 and 4, where 1 corresponds to the respondent's favorite algorithm for each testing condition; a certain audio type played at a certain speed. Since respondents were allowed to give two or more algorithms a tied ranking if they perceived them to be indistinguishable, data entries had to be manually corrected to represent a standardized format. We used the standard competition ranking method (or "1224 ranking") for handling ties. In some cases, participants had shifted the scale down, marking 2 or 3 as the highest rank. In those cases, the rankings were simply shifted up to start from rank 1 and then adjusted according to the competition ranking system. Examples of this process are shown in table 1.

Table 1: Examples of ranking standardization.

Algorithm	A	B	C	D
User ranking	3	2	3	4
Corrected ranking	2	1	2	4
User ranking	4	4	4	4
Corrected ranking	1	1	1	1
User ranking	2	2	2	3
Corrected ranking	1	1	1	4

Although the results are represented numerically, the numbers have to be treated like data points on an ordinal scale. An ordinal scale only indicates the order of items, not the distance between them [18, 19]. Therefore, the individual scale value of an algorithm is by itself void of information but gains meaning when compared to the rankings of the others. For ordinal scales, parametric methods such as mean and standard deviation are inappropriate as the values on which they would be based have no numerical significance [18]. Instead, we will use non-parametric statistical methods like the Friedman test [17], ART ANOVA [25], and Nemenyi-test [10].

The code for performing the Friedman, Nemenyi and ART ANOVA tests is provided on GitHub.⁵

3.4.1 Friedman Test. The Friedman test was used to analyze the variance in rankings for the four algorithms across the 16 combinations of audio type and playback speed. The test answers the question: *Within this specific audio type and speed, do participants rank the algorithms differently?* The Friedman test performs one-way repeated measures analysis of variance (ANOVA). “One-way” refers to one dependent variable, in this case a specific combination of audio type and speed, meaning it cannot help us draw conclusions about the effects of audio type and playback speed on algorithm rankings independently [17]. The Friedman test was performed using `scipy.stats.friedmanchisquare` in Python 3.10 [22].

3.4.2 Aligned Rank Transform (ART) ANOVA. Due to the fact that only one independent variable is allowed, the Friedman test lacks the ability to describe the individual effects of audio types and playback speeds on algorithm rankings. A factorial repeated measures ANOVA test is needed to analyze the effects of audio type and playback speed separately. Due to our ordinal-scale data, the method would also need to be non-parametric. The aligned rank transform allows us to use a parametric repeated measures ANOVA by first aligning the ranking data. This was done using ARTool in R. [25]

3.4.3 Post-Hoc Tests. A deeper understanding of patterns in algorithm preference was gained using the Nemenyi test, which performs pairwise comparisons of each algorithm included in the Friedman test [10]. The Nemenyi test thus provides insight into whether algorithms significantly outperform other algorithms, although only overall and not per condition.

3.5 Qualitative Data

Summary and analyses of qualitative data were performed according to the General Inductive Approach proposed by Thomas [20].

⁵Code for statistical tests: https://github.com/hanneswesterberg/dm128x_stats

The respondents’ descriptions of their listening experiences and their sonic criteria for TSM algorithms frequently involved the use of terminology that was informal or even incorrect or inaccurate given the context. We have attempted to interpret the free text responses in a way that captures their essence and intended meaning, and placed them into categories of themes which appeared in several of the responses. The themes and categories were defined with the evaluation objectives in mind, but the findings themselves stem from the raw data alone and not from previous expectations or presumptions.

4 Results

We found that there is significant conformity in the preferences of time-stretching algorithms, depending on the playback speed and audio type. The real-time algorithms OLA and SOLA performed significantly worse overall compared to the more computationally heavy *élastique* and phase vocoder algorithms in most conditions. One notable exception was the speech recording at 0.9x speed, where the SOLA algorithm performed best.

A total of 29 participants completed the survey. The largest age group was 18-24 (N = 15), followed by 25-34 (N = 10), 55-64 (N = 3) and 65+ (N = 1). 17 respondents were women and 12 were men. The following sections will provide details of the responses for background information and test environment, the rankings, and finally the results of the statistical analysis.

4.1 Playback Speed Habits

When asked about their previous experience with audio time stretching, 20 out of 29 respondents chose “None”, eight chose “A little, I have tried it when working with audio”, and one chose “Quite a bit, I know which algorithms are appropriate for different types of audio”. The fourth option, “I am an industry expert familiar with the coding and mathematics of algorithms”, was not chosen by any respondent. Of the nine respondents who indicated some level of previous experience, three were women and six were men.

Respondents were asked to rate how bothered they are by poor audio quality in speech and music, on a scale 1-5. The results are summarized in table 2.

Table 2: Participants’ self-reported disturbance by poor audio quality (1 = low, 5 = high).

	SPEECH			MUSIC		
	Total	Women	Men	Total	Women	Men
Average	3.48	3.65	3.25	4.00	4.18	3.75
Std. dev.	1.06	0.79	1.36	0.93	0.64	1.22

When asked if they ever changed the playback speed of media, 16 respondents chose “I sometimes speed up media content”, eight chose “Never”, and five chose “I sometimes speed up and sometimes slow down content”. No respondent chose the option “I sometimes slow down media content”. Six out of 12 male respondents indicated that they never change playback speed, compared to only two out of 17 female respondents.

Lastly, participants were asked to provide a brief description of their use cases to change media playback speed if they had any. 17 respondents indicated that they speed up content, such as podcasts or recorded lectures, when the presenter or teacher speaks too slowly. Other reasons for increasing the playback speed were due to the respondent feeling bored, impatient, or in a hurry. Reasons for reducing playback speed included making it easier to follow along in tutorials, having more time to take notes, learning difficult pieces of music, and learning dance choreographies.

4.2 Respondent Listening Focus

The respondents were asked to describe whether they had focused their listening on any particular aspect of the recording while examining the algorithms for each type of audio. Additionally, they were asked whether they considered speech or music more sensitive to the choice of TSM algorithm, after taking part in the test.

4.2.1 Classical Recording. After listening to the classical music recording, five respondents said they were unsure whether they had focused on something particular, while two said they simply focused on what sounded “best” or “most pleasing”. Eight respondents mentioned either a preference for “smoothness”, or a dislike of “choppiness”, “stutter”, or “vibrations”. We have chosen to interpret these as one consistent theme, as some of these comments were made in direct reference to the OLA algorithm, whose windowing becomes very audible at lower playback speeds. It is worth noting that the SOLA algorithm exhibits a similar behavior, so some of the comments may also be referred to it. Another respondent mentioned a dislike for an “eco/delay” effect, which may be a reference to the same windowing artifact. Additional listening criteria included “similarity to the original”, “clarity”, and “purity”. Some respondents mentioned a dislike for “distortion” and “brokenness”. Several respondents reported focusing on the quality of the high frequencies, although this may be a consequence of our question description containing a reference to spectral properties, which may not have occurred to the respondents to consider otherwise.

4.2.2 Speech Recording. For the speech recording, some respondents reported that they focused on specific words, parts of words, or sounds produced by the speaker. Three participants mentioned fricatives or s sounds. Others mentioned general spectral properties of the sound, with three mentioning a dislike of “metallic” or “tinny” sounds. Again, there were mentions of “eco/delay” effects, but this time also a “chorus” effect, which likely was in reference to the phase vocoder algorithm which can yield a smeared, phasy result when applied to monophonic sounds with transient content. The number of respondents who said they were uncertain or simply focused on what they liked best was the same as for the classical recording, although some of them were different respondents this time. “Clarity” was a popular listening criterion.

4.2.3 Instrumental Music Recording. After listening to the instrumental music recording, participants were asked which instruments they mainly focused on. They were allowed to select multiple answers. The results are summarized in table 3.

Table 3: Respondent focus on instruments when assessing algorithm quality for instrumental music.

Instrument	Chosen by (out of 29)
Lead guitar	19
Drums	13
Bass	8
Keys/piano	8
Everything equally	2
Other	4

Two of the respondents who chose “other” made reference to a trumpet which they believed was part of the mix. The song contained no trumpet playing, but there was a synthesizer doubled with the lead guitar playing the melody which may have caused the confusion. A third respondent specifically referenced the hi-hat cymbal of the drum kit, while the fourth made no reference to any particular instrument.

4.2.4 Music With Vocals Recording. Respondents were asked which instruments they focused on when listening to music with vocals. Results are summarized in table 4. In both the instrumental record-

Table 4: Respondent focus on instruments when assessing algorithm quality for music with vocals.

Instrument	Chosen by (out of 29)
Vocals	27
Guitar	16
Drums/percussion	4
Everything equally	1
Other	0

ing and the one with vocals, respondents appeared to focus mainly on the most upfront elements in the mix. The lead guitar in the instrumental song serves as the main vehicle of melody — just as the vocals do in the recording of music with singing.

4.2.5 Perceived Algorithm Sensitivity of Speech and Music. Prior to listening to and evaluating the audio examples, participants rated how bothered they were by poor audio quality in general for both speech and music (see table 2). After completing the test, participants were asked whether considered speech or music to be more sensitive to the choice of stretching algorithm. 17 participants chose “Speech”, six chose “Music”, and another six chose “Both”. Participants also rated the importance of the time-stretching algorithm for the overall listening experience on a scale from 1 (“not important”) to 5 (“crucial”). The median and mode ratings were both 4, with an average of 3.83 and a standard deviation of 1.04.

4.3 Algorithm Rankings

Tables 5-8 show the median and mode of algorithm rankings across all four playback speeds for each audio file (N=29).

Table 5: Median | mode of algorithm rankings for the classical music recording.

Speed	OLA	SOLA	Phase voc	Élastique
1.5x	4 4	2 1	1 1	2 3
0.9x	4 4	2 1	1 1	2 1
0.75x	4 4	3 3	1 1	2 2
0.5x	3 3	3 3	1 1	2 1

Table 6: Median | mode of algorithm rankings for the speech recording.

Speed	OLA	SOLA	Phase voc	Élastique
1.5x	3 3	1 1	4 4	2 2
0.9x	3 3	1 1	3 4	2 2
0.75x	3 4	3 2	3 4	1 1
0.5x	3 3	4 4	2 2	1 1

Table 7: Median | mode of algorithm rankings for the instrumental music recording.

Speed	OLA	SOLA	Phase voc	Élastique
1.5x	4 4	2 1	1 1	1 1
0.9x	4 4	3 3	1 1	1 1
0.75x	3 3	3 3	1 1	1 1
0.5x	4 4	3 3	1 1	1 1

Table 8: Median | mode of algorithm rankings for the recording of music with vocals.

Speed	OLA	SOLA	Phase voc	Élastique
1.5x	4 4	3 3	2 1	1 1
0.9x	4 4	1 1	2 2	1 1
0.75x	3 3	3 3	2 2	1 1
0.5x	3 3	4 4	2 2	1 1

4.4 Friedman Test

We conducted the Friedman test on the ranking data in order to determine statistical significance of ranking variance. Each algorithm was ranked 16 times by each participant, for all combinations of audio type and playback speed. All 16 combinations were treated as test conditions, with 29 data points for each. Having four different algorithms, the test was done with three degrees of freedom. We chose a significance level of 95% which, given three degrees of freedom, gives us a critical chi-square value of 7.815. The null hypothesis of the Friedman test is that the variance in ranks is caused by chance, which will be rejected for chi-square values above the critical value of 7.815. The results of the Friedman test are summarized in table 9. All tested conditions produced a chi-square value well above the critical, and a p-value far below 0.05. The null hypothesis was therefore rejected and further post-hoc tests were performed.

Table 9: Friedman statistics per combination of audio type and playback speed

Audio	Speed	Chi-square	p-value
Classical	0.5	58.904	1.008e-12
	0.75	66.896	1.971e-14
	0.9	34.305	1.708e-07
	1.5	37.533	3.550e-08
Instr. music	0.5	54.548	8.571e-12
	0.75	55.297	5.934e-12
	0.9	48.920	1.357e-10
	1.5	40.792	7.238e-09
Music w/ vocals	0.5	54.096	1.070e-11
	0.75	53.283	1.596e-11
	0.9	39.909	1.114e-08
	1.5	43.905	1.581e-09
Speech	0.5	53.599	1.366e-11
	0.75	39.583	1.306e-08
	0.9	52.185	2.735e-11
	1.5	48.032	2.097e-10

4.5 Aligned Rank Transform ANOVA

To capture individual effects of audio and playback speed on algorithm rankings, as well as their interactions, we performed an aligned rank transform analysis of variance as described by Wobbrock et al. [25]. Results are shown in table 10.

Table 10: Aligned Rank Transform ANOVA (Type III Wald F Tests with Kenward-Roger df)

Effect	F	Df	Df _{res}	Pr(>F)
<i>Main Effects</i>				
Audio	0.004	3	1764	0.9997
Speed	0.144	3	1764	0.9336
Algorithm	489.50	3	1764	$< 2 \times 10^{-16}$
<i>Two-Way Interactions</i>				
Audio:Speed	0.052	9	1764	0.9999
Audio:Algorithm	54.13	9	1764	$< 2 \times 10^{-16}$
Speed:Algorithm	54.50	9	1764	$< 2 \times 10^{-16}$
<i>Three-Way Interaction</i>				
Audio:Speed:Algorithm	8.47	27	1764	$< 2 \times 10^{-16}$

The ART ANOVA results indicate that algorithm is a significant main effect on rankings, along with the two-way interactions between algorithm–audio and algorithm–playback speed, and the three-way interaction between algorithm–audio–playback speed.

4.5.1 Main Effects. Algorithm being a significant main effect should be interpreted in the same manner as the results of the Friedman test: that there are clear preferences when it comes to algorithms. Audio and playback speed alone are not main effects, which is to be expected given that the rankings pertain to the algorithms — not the audio types nor the playback speeds.

4.5.2 Two-Way Interactions. Both two-way interactions involving algorithm (algorithm–audio and algorithm–playback speed) are shown to be highly significant. This should be interpreted as algorithm preferences being highly dependent on both audio type and playback speed, independently. The two-way interaction between audio and playback speed is not significant, which is to be expected for the same reason as why they are not significant main effects.

4.5.3 Three-Way Interaction. The three-way interaction between algorithm, audio, and playback speed also had a highly significant effect on rankings. This indicates that the way algorithm preference varies with playback speed varies itself across audio types—and vice versa.

4.6 Post-Hoc Tests

4.6.1 Nemenyi Test. After the Friedman test showed significant results, the Nemenyi test was performed to reveal how each algorithm performed compared to the others. Results are summarized in table 11. We determine the directionality of the difference in performance using the mean rankings for each algorithm, summarized in table 12.

Table 11: Nemenyi post-hoc test results with significant p-values in bold (< 0.05).

	Élastique	Phase voc	OLA
Phase voc	8.93×10^{-2}	–	–
OLA	1.43×10^{-13}	7.36×10^{-7}	–
SOLA	7.36×10^{-7}	1.68×10^{-2}	8.93×10^{-2}

Table 12: Mean rankings of algorithms (1 = best, 4 = worst).

Algorithm	Mean ranking
Élastique	1.57
Phase voc	1.96
SOLA	2.60
OLA	3.25

Using the information in tables 11 and 12, we see that both the Élastique and Phase vocoder algorithms perform significantly better than both the OLA and SOLA algorithms. The Nemenyi test does not show a significant difference between the Élastique and Phase vocoder algorithms, nor between the OLA and SOLA algorithms. These results do not account for specific effects on rankings of audio type and playback speed, but show overall algorithm preference.

4.6.2 Pairwise t-tests with Bonferroni Correction. After the aligned rank transform ANOVA test produced significant results, we performed pairwise t-tests with Bonferroni correction to determine which pairs of algorithms perform significantly different. Results are shown in table 13, and reveal that differences in performance between all algorithms are statistically significant. These results contrast with those of the Nemenyi test.

Table 13: Pairwise t-test with Bonferroni correction. All p-values significant (< 0.05).

	Élastique	Phase voc	OLA
Phase voc	9.7×10^{-11}	–	–
OLA	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	–
SOLA	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$

5 Discussion

With this study, we set out to investigate whether user preferences in algorithms for time-scale modification of audio signals tend to converge or diverge for different combinations of audio types and playback speeds. We chose this topic because the playback speed feature has become ubiquitous in digital media players and is frequently used daily by a large group of media consumers. Despite this, only a handful of interested people possess knowledge about the algorithms which perform the change in playback rate, even though they produce greatly varying results.

Our results indicate that there is a great deal of conformity in the preferences of TSM algorithms. The respondents tended to mostly agree on which algorithms performed the best and worst, regardless of playback speed and audio type. However, the specific playback speed and audio type determined which algorithms were the most preferred and least preferred, indicating that there is no universally best algorithm.

Looking at tables 10 and 12, we see that algorithms containing frequency-domain operations performed significantly better than time-domain algorithms. This is not unexpected, given that frequency-domain algorithms are not constrained by real-time requirements and can afford a higher computational cost. The élastique algorithm was the clear favorite overall, which was to be expected given its level of sophistication.

A key finding is that participants tended to agree even on deviations from typical ranking patterns. The SOLA algorithm was the clear favorite for the speech recording at 0.9x speed, and also performed well for the music with vocals recording at 0.9x speed.

An interesting observation from comparing the background responses to the end of form responses, was that participants seemed to change their mind on whether it was music or voice audio that was more important when trying to retain quality. In table 2 from section 4.1, we can see that music was ranked an average of 4 on the 1-5 disturbance scale, while speech was ranked an average of 3.48. This indicates that before answering the survey, participants believed that good audio quality was slightly more important for music than for speech. In contrast, the responses to the final survey page showed that the participants found voice recordings to be more sensitive to algorithm choice.

The common themes which appeared in the participants’ artifact perceptions aligned with the overall ranking patterns. Whenever a particular algorithm underperformed, comments pertaining to the artifacts of that algorithm appeared in the accompanying description of participants’ listening strategies. Several participants commented on the fluttering artifacts caused by the time-domain algorithms, especially OLA.

5.1 Method Discussion

The methods for data collection and statistical analysis used in this study could easily have been improved. Due to technical limitations in the Aidaform survey application, we were unable to randomly select the order in which audio types and playback speeds appeared, which may have contributed to learning effects. However, as previously stated, we did shuffle the order in which the algorithms appeared on each of the survey pages and used anonymous labeling.

Our use of the Friedman test was not appropriate given the data we had. The Friedman test is only appropriate given a single independent variable, when we had three. We circumvented this by reducing our independent variables down to one, by letting each combination of audio type and playback speed serve as separate testing conditions, leaving algorithm as the only independent variable. This step was not necessary using the aligned rank transform ANOVA test, which produced essentially the same results as the Friedman test in addition to more detailed information regarding two-way and three-way interactions.

As for the post-hoc tests, the Nemenyi test was less appropriate for the same reasons as the Friedman test. This time, results actually differed between our two post-hoc tests, and we are more inclined to trust the results of the pairwise t-tests with Bonferroni correction due to it being more appropriate for our dataset. The pairwise t-tests showed statistically significant differences in performance between all pairs of algorithms, whereas the Nemenyi test showed no significant difference between the phase vocoder and *élastique* algorithms, nor between the OLA and SOLA algorithms.

We wish to further justify our selection of playback speeds. The results of the first part of our survey showed that it was far more common for participants to increase playback speed than to decrease it, which was expected. Nonetheless, we opted for three speeds below 1.0x and only one above. This is because increasing playback rate is comparatively simple and produces fewer artifacts, as the algorithm has an abundance of material from which to subtract. In contrast, lowering playback rate requires interpolating audio where there is none, leading to more noticeable artifacts. As artifact perception was the main focus of the study, we felt it was more important to include multiple speeds below 1.0x rather than above. Moreover, adding additional increased speeds, such as 1.25x and 2.0x, would have made the survey intolerably long for participants.

5.2 Future Work

There are many endeavors which can further our understanding of human perception and TSM artifacts. For a start, future research could look to create more robust descriptions of the specific artifacts media consumers are most bothered by. Even further, researching and refining TSM techniques using a human perception-centered approach, rather than a technical one, could lead to interesting ways of making algorithms more computationally efficient by ignoring the correction of certain artifacts completely.

We are also aware of the limited scope of this study, especially with regards to our survey demographic. Therefore, we encourage future researchers to conduct similar studies with a larger and more diverse group of respondents, especially with varying levels of experience within sound editing and production.

6 Conclusion

Our results indicate that there is significant conformity in preferences for time-stretching algorithms for audio. The most preferred and least preferred algorithms depended on the type of audio and the playback speed. However, there were clear favorites overall. Algorithms making use of frequency-domain operations performed significantly better than time-domain algorithms, with the notable exception of small changes in playback rate for speech signals.

The high degree of conformity in subjective preferences for algorithms is supported by recurring themes regarding artifact perception. The choppy sound produced by phase discontinuities between synthesis frames in time-domain algorithms was particularly disliked by participants, and resulted in unfavorable rankings.

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